

EXTERNAL ADVERSARIAL AUDIT — CERTIFICATE

Block C — Adversarial Counterexample Search (Finite / Closed-Form Claims)

Subject:

Linear-Error Clique Partitions of Split Graphs (Erdos #81, Paper III), review version v0.9.5. Lean formalization explicitly OUT of scope.

Scope of this block:

C1 $\nu_3^*(H(p,q,d))=F$ by exact rational sandwich certificates + raw exact simplex; C2 margin (4.5) integer grid to $p=150$ + 200k random to $p=1e9$; C3 Lemmas 5.1/5.2/7.1 + Cor 5.3 with packing certificates and exact ν_3 ; C4 Prop 10.1 at true scale incl. $p=2304$; C5 sharpness family; C6 Appendix B exhaustive; C7 Lemma 4.1 by exact simplex on arbitrary profiles.

Evidence summary:

See results/*.txt — zero violations across ~2.55 million instances.

Primary results file:

results/c2_margin_results.txt SHA-256:

85e96f2c29d8a6895c50d3ed4c6f87a143ae5b62a3ead8c1494ec65af8c92848

VERDICT: PASS

Auditor: external adversarial audit agent (Claude, Anthropic model claude-fable-5), independent tooling.

Date: 2026-07-21 Package audited: see received_inputs.sha256